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DNA TEST REPORT - MEDGENOME LABS

Full Name / Ref No:	**SHIVANSHU GUPTA**	Order ID/Sample ID:	**527250/7739815**
Gender:	Male	Sample Type:	Blood
Date of Birth / Age:	19 years	Date of Sample Collection:	3rd November 2022
Referring Clinician:	Dr. Vishnu V. Y., AIIMS, New Delhi	Date of Sample Receipt:	4th November 2022
		Date of Order Booking:	5th November 2022
		Date of Report:	2nd December 2022
Test Requested:	**Whole Exome Sequencing**		

CLINICAL DIAGNOSIS / SYMPTOMS / HISTORY

Mr. *Shivanshu Gupta*, presented with clinical indications of delayed motor milestones, progressive gait difficulties, bilateral proximal upper and lower limb weakness, areflexia, scoliosis, decreased sensation in bilateral feet and pes cavus. Examination revealed decreased touch and pain sensation and lower motor neuron palsy. EPS test revealed sensorimotor demyelinating polyneuropathy with second degree axonal changes. EMG revealed signs of reinnervation. There is a positive family history in his elder sister. Mr. *Shivanshu Gupta* is suspected to be affected with Charcot-Marie-Tooth disease and has been evaluated for pathogenic variations.

RESULTS

PATHOGENIC VARIANT CAUSATIVE OF THE REPORTED PHENOTYPE WAS DETECTED

SNV(s)/INDELS

Gene# (Transcript)	Location	Variant	Zygosity	Disease (OMIM)	Inheritance	Classification$
***SH3TC2* (-)** (ENST00000515425.6)	Exon 14	c.3325C>T **(p.Arg1109Ter)**	Homozygous	Charcot-Marie-Tooth disease type 4C (OMIM#601596)	Autosomal recessive	Pathogenic

Copy Number Variants CNV(s)

No significant CNVs for the given clinical indications that warrants to be reported was detected.

NOTE:

- No other SNV(s)/INDELS or CNV(s) that warrants to be reported was detected. All the genes covered in this assay have been screened for the given clinical indications. To view the coverage of all genes Click here. NGS test methodology details of this assay are given in the appendix.
- $Genetic test results are reported based on the recommendations of American College of Medical Genetics [PMID: 25741868].

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- Please contact genetic.counseling@medgenome.com for genetic counselling. For any further technical queries please contact techsupport@medgenome.com

VARIANT INTERPRETATION AND CLINICAL CORRELATION

Variant description: A homozygous nonsense variation in exon 14 of the ***SH3TC2*** gene **(chr5:g.149010272G>A; Depth: 111x)** that results in a stop codon and premature truncation of the protein at codon 1109 **(p.Arg1109Ter; ENST00000515425.6)** was detected (Table). The observed variation has previously been reported in patients affected with Charcot-Marie-Tooth disease [PMID:16326826][ClinVar]. The p.Arg1109Ter variant has not been reported in the 1000 genomes and gnomAD databases and has a minor allele frequency of 0.06% in our internal database. The *in silico* prediction[#] of the variant is damaging by MutationTaster2. The reference codon is conserved across species.

OMIM phenotype: Charcot-Marie-Tooth disease type 4C (OMIM#601596) is caused by homozygous or compound heterozygous mutations in the *SH3TC2* gene (OMIM*608206). This disorder is characterized by progressive weakness and atrophy, initially of the peroneal muscles and later of the distal muscles of the arms. Charcot-Marie-Tooth disease is classified in two main groups on the basis of electrophysiologic properties and histopathology: primary peripheral demyelinating neuropathies and primary peripheral axonal neuropathies. Demyelinating neuropathies are characterized by severely reduced nerve conduction velocities (less than 38 m/sec), segmental demyelination and remyelination with onion bulb formations on nerve biopsy, slowly progressive distal muscle atrophy and weakness, absent deep tendon reflexes, and hollow feet. Demyelinating Charcot-Marie-Tooth disease is characterized by onset in childhood, early-onset scoliosis and a distinct Schwann cell pathology. [PMID: 18842627].

Based on the above evidence[$], ***this SH3TC2 variation is classified as a pathogenic variant and has to be carefully correlated with the clinical symptoms.***

The significance/classification of the variants may change based on the genetic testing in parents and other family members.

RECOMMENDATIONS

- Sequencing the variants in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Genetic counselling is advised for interpretation on the consequences of the variants.
- If results obtained do not match the clinical findings, additional testing should be considered as per referring clinician's recommendations.
- The sensitivity of NGS assay to detect copy number variations (CNV) is 70-75%. we recommend discussing alternative testing methodology option with MedGenome Tech Support as required.

Sandhya Nair, Ph.D Sr. Manager - Variant Interpretation	Balaji Rajashekar, Ph.D Director - Clinical Bioinformatics	Dr. Meenakshi Lallar, MS (Obgyn), DM (Medical Genetics) Consultant - Clinical Geneticist

APPENDIX

TEST METHODOLOGY

Targeted gene sequencing: Selective capture and sequencing of the protein coding regions and clinical relevant in the genome is performed. Variants identified in the exonic regions and splice-site are generally actionable compared to variations that occur in non-coding regions. Targeted sequencing represents a cost-effective approach to detect variants present in multiple/large genes in an individual.

DNA extracted from blood was used to perform targeted gene capture using a custom capture kit. The libraries were sequenced to mean depth of >80-100X on Illumina sequencing platform. We follow the GATK best practices framework for identification of germline variants in the sample using Sentieon [Sentieon]. The sequences obtained are aligned to human reference genome (GRCh38) using BWA aligner [Sentieon, PMID:20080505] and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels [Sentieon]. Sentieon haplotype caller is then used to identify variants in the sample. The germline variants identified in the sample is deeply annotated using VariMAT pipeline. Gene annotation of the variants is performed using VEP program [PMID: 20562413] against the Ensembl release 99 human gene model [PMID: 29155950]. In addition to SNVs and small Indels, copy number variants (CNVs) are detected from targeted sequence data using the ExomeDepth method PMID: 22942019]. This algorithm detects CNVs based on comparison of the read-depths in the sample of interest with the matched aggregate reference dataset.

Clinically relevant mutations in both coding and non-coding regions are annotated using published variants in literature and a set of diseases databases : ClinVar, OMIM, HGMD, LOVD, DECIPHER (population CNV) and SwissVar [PMID: 26582918, 18842627, 28349240, 21520333, 19344873, 20106818]. Common variants are filtered based on allele frequency in 1000Genome Phase 3, gnomAD (v3.1 & 2.1.1), dbSNP (GCF_000001405.38), 1000 Japanese Genome, TOPMed (Freeze_8), Genome Asia, and our internal Indian population database (MedVarDb v2.1) [PMID: 26432245, 32461613, 11125122, 26292667, 33568819, 31802016]. Non-synonymous variants effect is calculated using multiple algorithms such as PolyPhen-2, SIFT, MutationTaster2 and LRT. Clinically significant variants are used for interpretation and reporting.

Average sequencing depth (x)	Average on-target sequencing depth (x)	Percentage target base pairs covered		
		0x	≥5x	≥20x
277	123.43	1.30	97.71	95.34

Total data generated (Gb)	10.18
Total reads aligned (%)	99.98
Reads that passed alignment (%)	91.17
Data ≥Q30 (%)	96.08

$The classification of the variations is done based on American College of Medical Genetics as described below [PMID: 25741868].

Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
Pathogenic	A disease-causing variation in a gene which can explain the patient's symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.

Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

[#]The transcript used for clinical reporting generally represents the canonical transcript (MANE Select), which is usually the longest coding transcript with strong/multiple supporting evidence. However, clinically relevant variants annotated in alternate complete coding transcripts could also be reported.

Variants annotated on incomplete and nonsense mediated decay transcripts will not be reported.

[#]The *in-silico* predictions are based on Variant Effect Predictor (v104), [SIFT version - 5.2.2; PolyPhen - 2.2.2; LRT version (November, 2009); CADD (v1.6); Splice AI; dbNSFPv4.2] and MutationTaster2 predictions are based on NCBI/Ensembl 66 build (GRCh38 genomic coordinates are converted to hg19 using UCSC LiftOver and mapped to MT2).

Diseases databases used for annotation includes ClinVar (updated on 5082021), OMIM (updated on 5082021), HGMD (v2021.3), LOVD (Nov-18), DECIPHER (population CNV) and SwissVar.

LIMITATIONS

- Genetic testing is an important part of the diagnostic process. However, genetic tests may not always give a definitive answer. In some cases, testing may not identify a genetic variant even though one exists. This may be due to limitations in current medical knowledge or testing technology. Accurate interpretation of test results may require knowing the true biological relationships in a family. Failing to accurately state the biological relationships in {my/my child's} family may result in incorrect interpretation of results, incorrect diagnoses, and/or inconclusive test results.
- Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete.
- Specific events like copy number variations, translocations, repeat expansions and chromosomal rearrangements may not be reliably detected with whole exome sequencing. Variants in untranslated region, promoters and intronic variants are not assessed using this method.
- Genetic testing is highly accurate. Rarely, inaccurate results may occur for various reasons. These reasons include, but are not limited to: mislabelled samples, inaccurate reporting of clinical/medical information, rare technical errors or unusual circumstances such as bone marrow transplantation, blood transfusion; or the presence of change(s) in such a small percentage of cells that may not be detectable by the test (mosaicism).
- The variant population allele frequencies and *in silico* predictions for GRCh38 version of the Human genome is obtained after lifting over the coordinates from hg19 genome build. The existing population allele frequencies (1000Genome, ExAC, gnomAD-Exome) are currently available for hg19 genome version only. This might result in some discrepancies in variant annotation due to the complex changes in some regions of the genome.

DISCLAIMER

- Interpretation of variants in this report is performed to the best knowledge of the laboratory based on the information available at the time of reporting. The classification of variants can change over time and MedGenome cannot be held responsible for this. Please feel free to contact MedGenome Labs (techsupport@medgenome.com) in the future to determine if there have been any changes in the classification of any variations. Re-analysis of variants in previously issued reports in light of new evidence is not routinely performed but may be available upon request.
- The sensitivity of this assay to detect large deletions/duplications of more than 10 bp or copy number variations (CNV) is 70-75%. The CNVs detected have to be confirmed by alternate method.
- Due to inherent technology limitations of the assay, not all bases of the exome can be covered by this test. Accordingly, variants in regions of insufficient coverage may not be identified and/or interpreted. Therefore, it is possible that pathogenic variants are present in one or more of the genes analysed but have not been detected. The variants not detected by the assay that was performed may impact the phenotype.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- The mutations have not been validated/confirmed by Sanger sequencing.
- Incidental or secondary findings (if any) that meet the ACMG guidelines [PMID: 27854360] can be given upon request.
- The report shall be generated within turnaround time (TAT), however, such TAT may vary depending upon the complexity of test(s) requested. MedGenome under no circumstances will be liable for any delay beyond afore mentioned TAT.
- It is hereby clarified that the report(s) generated from the test(s) do not provide any diagnosis or opinion or recommends any cure in any manner. MedGenome hereby recommends the patient and/or the guardians of the patients, as the case may be, to take assistance of the clinician or a certified physician or doctor, to interpret the report(s) thus generated. MedGenome hereby disclaims all liability arising in connection with the report(s).
- In a very few cases genetic test may not show the correct results, e.g., because of the quality of the material provided to MedGenome. In case where any test provided by MedGenome fails for unforeseeable or unknown reasons that cannot be influenced by MedGenome in advance, MedGenome shall not be responsible for the incomplete, potentially misleading or even wrong result of any testing if such could not be recognised by MedGenome in advance.
- This is a laboratory developed test and the development and the performance characteristics of this test was determined by MedGenome.

END OF REPORT